[2]

9 Lee started work on 1 January 2019.
He started with a monthly salary of $4100 and has seen his salary increase by 4% annually.

(a) Show that Lee's current monthly salary, in January 2024, is $5000, correct to the nearest thousand.

Answer

[1]

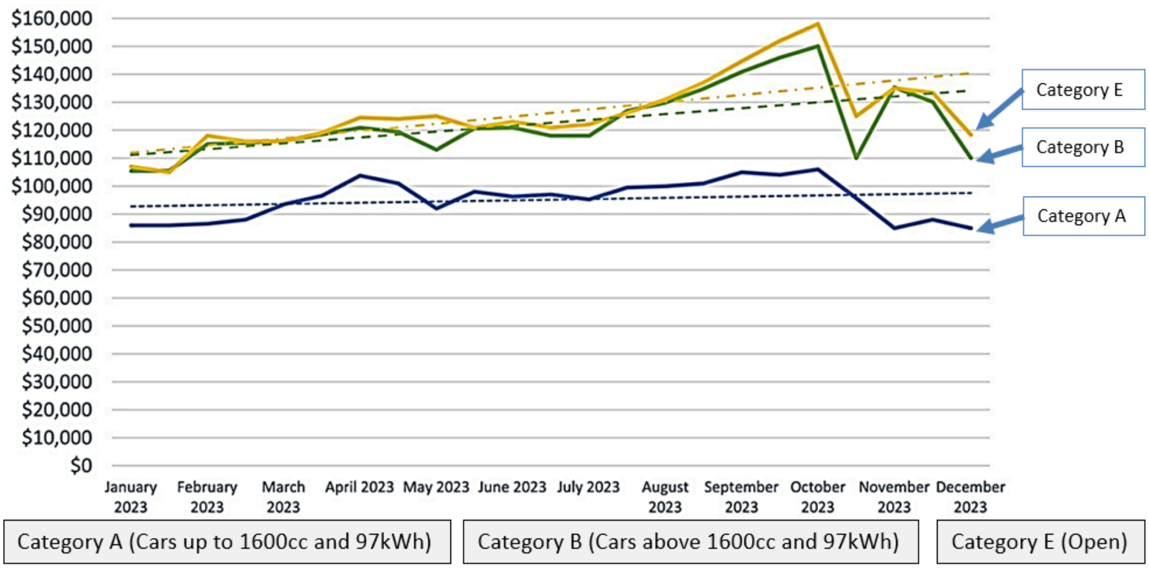
Lee has savings of $105 000.
With his savings and monthly income, he intends to buy a car, in January 2024.
He is deciding between buying an EV 60kWh electric car or a 1998cc petrol car.

The table below is used to calculate the cost price of the car:

	Car Type	EV 60kWh electric car	1998cc petrol car
a.	Base cost (without COE)	$100,000	$110,000
b.	COE	Refer to Table A (COE Prices)	
c.	Rebates (clean energy initiatives)	$30,000	Not applicable
Total cost price of car = a + b – c			

Every car owner in Singapore must purchase a Certificate of Entitlement (COE) for the car, which gives him the right to own and use a vehicle in Singapore. The COE prices since January 2023 can be seen below.

Table A: COE Prices



Lee will take a loan from a financial institution. He intends to take the largest loan possible. The loan amount can be calculated using the information in Table B below:

Table B: Calculation of Loan

Car Type	EV 60kWh	1998cc petrol car
Maximum loan amount	60% of cost price of car	70% of cost price of car
Maximum loan period	7 years	7 years
Interest rate (based on simple interest)	2.78% yearly	2.78% yearly

Being prudent, he would like to maintain an amount equivalent to at least 6 months of his salary in his savings.

(b) Show that Lee can only make the downpayment for **one** of the two cars.
Show your calculations clearly and justify any decisions you make.

Answer

[3]